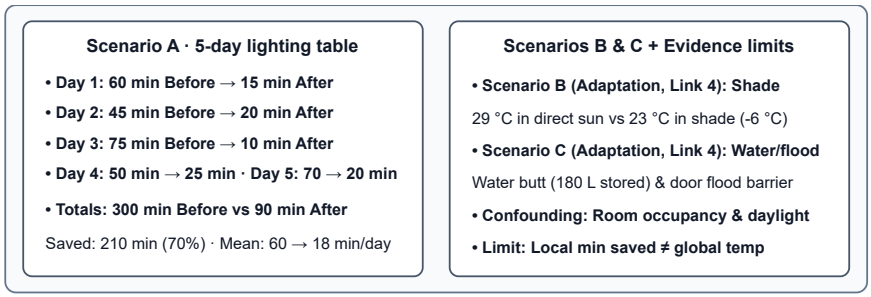
Reference sheets

Reference A · Safe Climate Action Scenarios (Scenarios A, B, and C) and 5-Day Practice Data

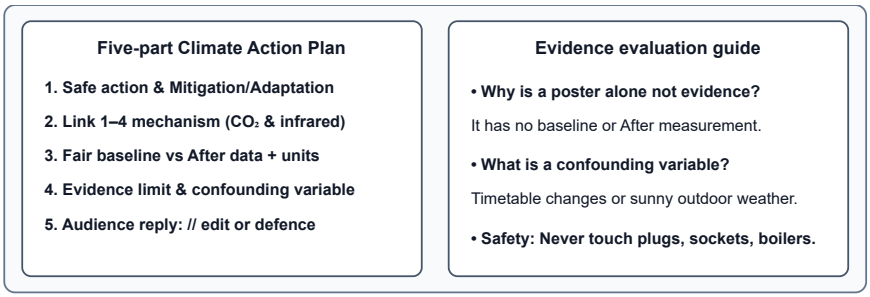


Five-day light table (300 min Before vs 90 min After). It also shows shade data (29 °C vs 23 °C) and data limits.

Reference only. Do not write on this sheet. All scenarios are safe paper datasets. Pupils never touch plugs, sockets, boilers, or live controls.

Scenario code.	Action and chain type.	5-day Before baseline vs 5-day After data.	Key Stage 3 mechanism and data limit.
Scenario A · Unused lights routine.	Turn off lights when room is empty. Mitigation (Links 1–3).	Days 1–5 Before: 60, 45, 75, 50, 70 min (300 min; 60 min/day). After: 15, 20, 10, 25, 20 min (90 min; 18 min/day). 210 min (70%) saved.	Cuts power demand (kWh), fossil fuel burning, and CO_2 infrared heat. Limit: measures room time, not kg CO_2 or global climate.
Scenario B · Shaded cooling space.	High-albedo light roof and shade in yard. Adaptation (Link 4).	5-day midday thermometer mean: 29 °C in direct sun. 23 °C under shaded roof (6 °C cooler).	Light roof (higher albedo) reflects sun rays and cuts heat at Link 4. Limit: does not cut CO_2 release at Link 1.
Scenario C · Water butt & flood gate.	Store roof rain and fit doorway flood gate. Adaptation (Link 4).	Water butt stores 180 L for dry days. Door gate keeps floor dry in a 25 mm rain test.	Cuts harm from drought and flood water at Link 4. Limit: does not remove greenhouse gases from the air.

Reference B · Five-part Climate Action Plan and fair-test guide



Five-part Climate Action Plan drawing. It shows safe action, mechanism, baseline vs after data, limits, and reply.

Plan part.	What to include in your plan.	Example from Scenario A (Unused lights routine).
1. Action & type.	Name the safe scenario. State Mitigation (Link 1) or Adaptation (Link 4).	'Scenario A (Unused lights routine) is mitigation at Link 1 and Link 2.'
2. Science mechanism.	Link the action to fossil fuels, CO_2, infrared heat, or Link 4 harm.	'Using less power cuts fossil fuel burning at power plants. That adds less CO_2 greenhouse gas to trap outgoing infrared heat.'
3. Baseline vs After data.	Compare Before baseline and After totals and means with units.	'Over 5 days, light waste fell from 300 minutes (60 min/day) Before to 90 minutes (18 min/day) After. That saved 210 minutes (70%).'
4. Limit & confounding factor.	State one confounding factor and what the data cannot prove.	'Confounding factor: room timetable changes or sunny weather. Limit: 5 days of room minutes cannot prove global climate cooling.'
5. Audience // reply.	Answer a review question and fix any over-claim with //.	'// Saved 210 minutes of wasted room light (Link 1 mitigation). It did not stop global warming in 5 days.'

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Steps 1 and 2 of the arrival drawing. Light waste fell from 60 min Before to 15 min After on Day 1. How many minutes were saved?

Help. Take 15 away from 60 and write the unit (minutes).

2. Look at Step 3 of the arrival drawing. Across 5 days, light waste fell from 300 minutes to 90 minutes. How many minutes were saved?

Help. Take the 5-day After total (90) from the 5-day Before total (300).

3. The class timed the empty room for 5 days Before the action (the baseline). Why did they not only time After?

Help. Think about what number we compare the After score against.

4. How does cutting wasted power in buildings link to Links 1 and 2?

Help. Connect power demand to fossil fuel burning and carbon dioxide.

You do tasks

1. Use Reference A and Reference B. Write a Climate Action Plan for Scenario A (Unused lights routine). (a) Sort it as mitigation or adaptation and explain its Link 1–3 mechanism. (b) Compare the 5-day Before baseline (300 min) with the 5-day After data (90 min).

Staff I ☐ S ☐ C ☐

Help. Include the 210 minutes saved. Link power use to fossil fuel burning and carbon dioxide.

2. Compare Scenario A (Unused lights) with Scenario B (Shade: 23 °C vs 29 °C). Explain why a town needs both kinds of action. **Staff I ☐ S ☐ C ☐**

Help. Contrast Link 1 emission cuts (mitigation) with Link 4 heat care (adaptation).

3. Use Reference B. State one limit of the 5-day light data and one confounding factor. Then write a // fix for the claim: 'Our 5-day switch-off routine stopped global warming.'

Staff I ☐ S ☐ C ☐

Help. Check Part 4 and Part 5 on Reference B.

Optional participation-loop prompt. Share one part of your climate-action plan with an adult. Hear one review question or feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Look at Scenario A (Unused lights: 300 min Before to 90 min After). How does it link to the four-link warming chain?

2. Look at Scenario B (Shade: 29 °C in sun vs 23 °C in shade). Why is it sorted as adaptation rather than mitigation?

3. Give one limit of our 5-day room light data.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt